Jinwoo Jung

M.S. student, Hanyang University | ASML Lab, advised by Prof. Joon-Hyuk Chang

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Research Interests

Diffusion and flow-based generative models; controllable generation and editing; training-free inference efficiency.

Education

Hanyang University, Seoul, Republic of Korea — Mar 2025 – Feb 2027 (expected)
M.S. in Artificial Intelligence Semiconductor Engineering
- GPA 4.44/4.50.

Konkuk University, Seoul, Republic of Korea — Mar 2019 – Feb 2025
B.S. in Computer Science and Engineering
- GPA 3.88/4.50 (major 3.97/4.50).

Zurich University of Applied Sciences (ZHAW), Winterthur, Switzerland — Sep 2022 – Jan 2023
Exchange student, School of Engineering

Publications

* denotes equal contribution; † denotes corresponding author.

SARA: Step-Adaptive Rank Adjustment for Diffusion Inference Acceleration — NeurIPS 2026
Dong Hoon Lim*, **Jinwoo Jung***, Won-Gook Choi, Joon-Hyuk Chang† [paper] [code]

Edit the Moment, Keep the Rest: Time-Localized Audio Editing via Instruction — INTERSPEECH 2026
Jinwoo Jung, Gihun Son, Won-Gook Choi, Joon-Hyuk Chang† [paper] [demo]

Branch-wise Complementary Attention for Acoustic Scene Classification — INTERSPEECH 2026
Seung-Gyu Han*, **Jinwoo Jung***, Pil Moo Byun, Won-Gook Choi, Joon-Hyuk Chang† [paper]

Machine Learning Algorithm-Based Prediction Models for Adolescent Suicide Attempt — IEEE ICCE 2025
Jinwoo Jung, Seulmi Kang, Myeonghwan Go, Sieon Lee, Seokheon Cho† [paper]

Manuscripts

ENCORE: Training-Free Few-Step Diffusion Sampling with Terminal Moment Matching — under review, 2026
Jinwoo Jung, Dong Hoon Lim, Joon-Hyuk Chang†, Defang Chen

TRACE: Trajectory-Conditioned Editing of Sound Sources in First-Order Ambisonics — in preparation, 2026
Jinwoo Jung, Dong Hoon Lim, Joon-Hyuk Chang†

Presentations

NeurIPS 2026, Atlanta, USA — SARA (poster, presenting author) — Dec 2026 (upcoming)
INTERSPEECH 2026, Sydney, Australia — EMKR (poster, presenting author) and BCA (poster) — Sep 2026

Patents

All applications are filed in the Republic of Korea.

System and Method for Accelerating Diffusion Model Inference Using Step-Adaptive Low-Rank Adjustment — filed Oct 2026

Branch-wise Complementary Attention for Acoustic Scene Classification — filed Oct 2026

Interval Conditioning and Source-Event Masking for Instruction-based Time-Localized Audio Editing — filing in progress

Research Experience

Hanyang University, Seoul, Republic of Korea — Mar 2025 – present
ASML Lab (Acoustic, Speech Signal Processing and Machine Learning)
- AI Model Compression and Acceleration, with Samsung Electronics (Jan 2026 – present).
- Sound-Tracking AI Agent via Connected Devices, with LG Electronics (Jun 2025 – Sep 2025).

Qualcomm Institute, UC San Diego, La Jolla, United States — Jan 2024 – Feb 2024
Artificial Intelligence (AI) Development Project
- Team lead; our team placed first.
- The team project continued after the program and became a conference paper (IEEE ICCE 2025).

Awards

1st Place, IC-PBL+ Research Showcase, Hanyang University — Jun 2026
Outstanding Achievement Award, Qualcomm Institute AI Development Project, UC San Diego — Feb 2024

Skills and Languages

Programming Python, PyTorch, librosa, torchaudio | **Languages** Korean (native), English (TOEFL iBT 100)